PROMETHAZINE

Not <2 yr (respiratory depression)

IV tissue injury

CHEMICAL NAME (RS)-N,N-dimethyl-1-(10H-phenothiazin-10-yl)propan-2-amine (hydrochloride or teoclate)

CLASS Phenothiazine H1 antihistamine (sedating) with antiemetic action

TRADE NAMES Phenergan · Avomine (teoclate) · Sominex

THERAPEUTIC USES / INDICATIONS

- Nausea and vomiting (motion sickness, post-operative, hyperemesis gravidarum).
- Allergic reactions, urticaria, pruritus.
- Sedation (pre-operative, short-term insomnia).
- Adjunct in anaphylaxis (after adrenaline).

ADMINISTRATION

- **PO** tablets/elixir.
- **Deep IM** preferred; **IV** only diluted, slowly, into a large-bore running line — severe tissue injury/gangrene if extravasated or intra-arterial.
- Motion sickness: 1–2 h before travel.

DOSAGES

- **Nausea (adult):** 12.5–25 mg every 4–6 h; max 100 mg/day.
- **Motion sickness:** 25 mg 1–2 h before travel (teoclate 25 mg).
- **Allergy:** 10–20 mg 2–3 times daily, or 25 mg at night.
- **Children 2–5 yr:** 5–15 mg daily; 5–10 yr: 10–25 mg daily.

PHARMACOKINETICS

- **Onset:** PO 20 min; IM 20 min. **Duration:** 4–12 h.
- Metabolised in liver.
- Excreted in urine and bile. **t½:** 10–19 h.

PHARMACODYNAMICS

- **H1 antagonist** + D2 antagonism at CTZ + strong anticholinergic → antiemetic, sedative, antiallergic.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Marked drowsiness, confusion.
- Dry mouth, blurred vision.
- Urinary retention.

ADVERSE EFFECTS (SERIOUS)

- Fatal respiratory depression in children <2 yr.
- Severe tissue injury with IV/intra-arterial injection.
- Extrapyramidal effects; neuroleptic malignant syndrome.
- Paradoxical excitation in children.

CONTRAINDICATIONS & CAUTIONS

- Children <2 years.
- Coma, CNS depression.
- MAOIs within 14 days.
- Intra-arterial or SC injection.

KEY INTERACTIONS

- **Opioids, alcohol, benzodiazepines:** additive respiratory/CNS depression.
- **Anticholinergics:** ↑ effects.
- **MAOIs:** avoid.

NURSING MANAGEMENT

- Prefer deep IM; if IV, dilute, slow, check site; stop if pain/burning.
- Fall precautions; monitor sedation and RR.
- Check age before giving to children.

PATIENT EDUCATION

- Causes drowsiness — no driving or alcohol.
- Do not give to children under 2.
- Report burning pain at injection site.

KEY POINTS

- Sedating antihistamine-antiemetic.
- Not <2 yr; fatal respiratory depression.
- Used in hyperemesis gravidarum.
- IV → tissue necrosis; deep IM preferred.